CroBIM-U: Uncertainty-Driven Referring Remote Sensing Image Segmentation

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***Abstract*—Referring remote sensing image segmentation (RRSIS) aims to localize specific targets described by natural language within complex overhead imagery. However, due to extreme scale variations, dense similar distractors, and intricate boundary structures, the reliability of cross-modal alignment exhibits significant spatial nonuniformity. Existing methods typically employ uniform fusion and refinement strategies across the entire image, which often introduces unnecessary linguistic perturbations in visually clear regions while failing to provide sufficient disambiguation in confused areas. To address this, we propose an uncertainty-guided framework that explicitly leverages a pixel-wise referring uncertainty map as a spatial before orchestrating adaptive inference. Specifically, we introduce a plug-and-play referring uncertainty scorer (RUS), which is trained via an online error-consistency supervision strategy to interpretably predict the spatial distribution of referential ambiguity. Building on this prior, we design two plug-and-play modules: 1) uncertainty-gated fusion (UGF), which dynamically modulates language injection strength to enhance constraints in high-uncertainty regions while suppressing noise in low-uncertainty ones; and 2) uncertainty-driven local refinement (UDLR), which utilizes uncertainty-derived soft masks to focus refinement on error-prone boundaries and fine details. Extensive experiments demonstrate that our method functions as a unified, plug-and-play solution that significantly improves robustness and geometric fidelity in complex remote sensing scenes without altering the backbone architecture.**

***Index Terms*—Cross-modal fusion, hyperspectral image, plug-and-play module, referring remote sensing image segmentation (RRSIS), spatial nonuniformity, uncertainty estimation.**

I. Introduction

REMOTE sensing imagery, characterized by broad spatial coverage and frequent revisits, is fundamental to applications such as urban mapping, disaster assessment, ecological monitoring, and resource inventory [1], [2], [3]. In many real-world scenarios, however, users require query flexibility beyond predefined semantic categories (e.g., building or road). Instead, they often specify targets via natural-language descriptions of attributes, spatial relations, and context (e.g., the slender bridge crossing the river next to the port). This demand has motivated the emergence of referring remote sensing image segmentation (RRSIS) [1], [4]: given an image and a text description, the model predicts a pixel-wise mask of the referred instance. Compared with category-driven semantic segmentation, RRSIS enables more flexible human–computer interaction and supports instance-level, fine-grained mapping over large-scale geospatial data.

Compared with referring segmentation in natural scenes, referring image segmentation (RIS) [5], remote-sensing settings pose severe and unique challenges in cross-modal localization and boundary delineation [1], [6]. First, remote-sensing scenes are often dense and repetitive; visually similar instances (e.g., rooftops, ships, road fragments, and farmland parcels) can easily cause referential ambiguity [4], making cross-modal alignment reliability highly spatially nonuniform. Second, targets vary drastically in scale and orientation [1], [4], and the overhead viewpoint weakens viewpoint-dependent cues (e.g., perspective), increasing confusion under compositional descriptions. Third, many targets exhibit slender, fragmented, or low-contrast structures (e.g., roads, coastlines, dikes, and building contours), where errors tend to concentrate around boundaries [7]. These observations suggest that, even within a single image, both referential reliability (whether the correct instance is localized) and pixel prediction reliability (whether local geometry is stable) can vary substantially across space. Consequently, applying uniform fusion strength and refinement computation across all regions—as is common in current practices—may under-constrain high-risk locations while unnecessarily perturbing already reliable ones.

Existing approaches typically follow a two-tower design that encodes the image and text separately, predicting masks through cross-modal interaction in a decoder [1], [5]. While recent works have introduced multiscale interactions [4], foundation model priors (e.g., CLIP-SAM-based methods) [6], [8], or even generative diffusion models [9] to handle scale and semantic gaps, many implementations still lack an explicit, error-aligned spatial prior. Furthermore, although advanced generative foundation models [10] have improved visual representation, efficiently adapting them to pixel-level referring tasks with limited spatial selectivity remains challenging. In RRSIS, a uniform fusion strategy can induce two coupled issues. In visually confident regions with stable predictions, overly aggressive language modulation may introduce unnecessary perturbations. Conversely, in high-risk regions with severe ambiguity or fragile geometry (especially boundaries and thin structures), uniform fusion capacity can be insufficient for disambiguation and correction. This motivates risk-aware

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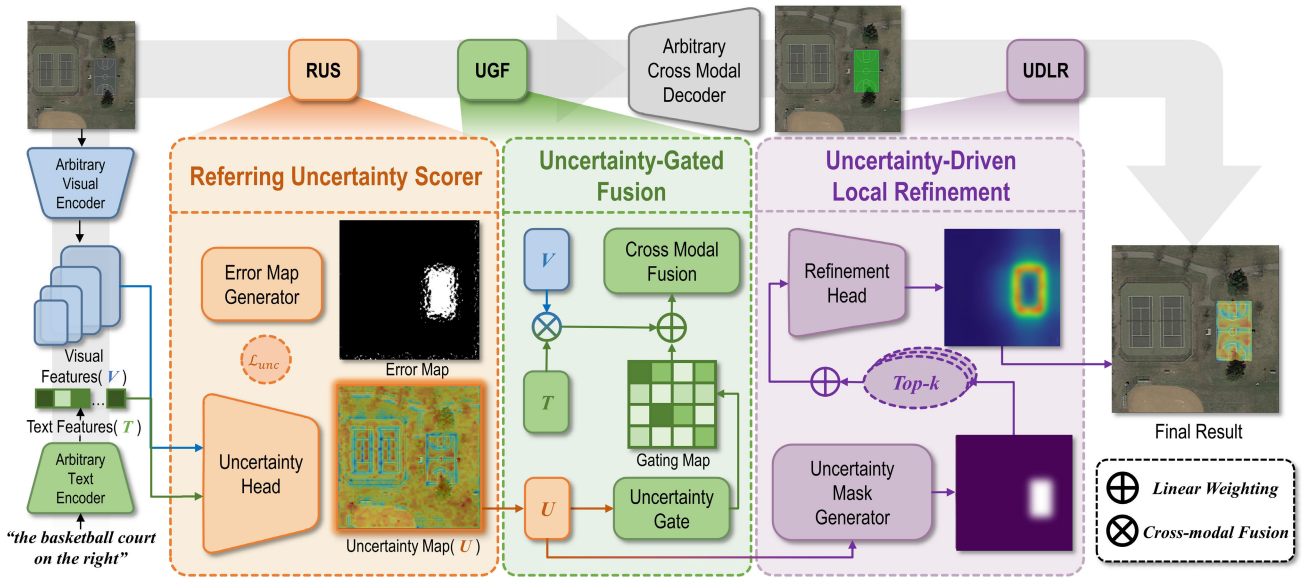


Fig. 1. Flowchart of the proposed risk-aware remote-sensing referring segmentation framework. Given an image I and a referring expression S, an arbitrary visual encoder and text encoder extract visual features $\mathbf{V}$ and text features $\mathbf{T}$. The RUS predicts a pixel-level grounding uncertainty map U (logits), which is supervised during training by an online error map generated from the decoder prediction, yielding the uncertainty loss $\mathcal{L}_{\text{unc}}$ (the error-map branch is used only in training). The predicted U provides a spatial before: 1) UGF, which performs a one-shot uncertainty-conditioned postfusion to modulate language injection and produce fused features for an arbitrary cross-modal decoder; and 2) UDLR, which converts U into a soft refinement mask to guide a lightweight refinement head for boundary and fine-structure correction, producing the final segmentation. $\oplus$ denotes linear weighting and $\otimes$ denotes cross-modal fusion.

modeling: beyond predicting what to segment, the model should estimate where mistakes are more likely and allocate cross-modal constraints and refinement resources accordingly.

To this end, we introduce a pixel-wise cross-modal referential uncertainty map as an intermediate representation of spatial risk. Here, uncertainty serves as a risk cue indicating that a location is more likely to suffer referential confusion or local segmentation error. During inference, the uncertainty map acts as a feed-forward spatial before enabling selective cross-modal fusion and focused local correction: language constraints are strengthened at high-uncertainty locations to resolve ambiguity, while unnecessary cross-modal perturbations are suppressed at low-uncertainty locations to preserve stable predictions. Moreover, the uncertainty prior concentrates refinement on error-prone areas such as boundaries and thin structures, improving geometric fidelity and mask consistency.

A key requirement is that the predicted uncertainty should be interpretable and semantically aligned with actual errors. Network design alone does not guarantee that uncertainty correlates with mistakes, especially under the complex ambiguities of remote-sensing scenes. We, therefore, propose an online risk learning strategy: during training, we derive a per-pixel error signal from the discrepancy between the model's current prediction and the ground truth (GT) and supervise uncertainty to be positively correlated with this error signal. This online, behavior-adaptive supervision unifies cross-modal disambiguation difficulty and boundary/detail instability into a single spatial prior that consistently guides fusion control and local correction.

Building on the uncertainty prior, we develop a risk-guided fusion–local correction pipeline with three plug-and-play modules. First, the referring uncertainty scorer (RUS) predicts the pixel-wise uncertainty map on top of encoder features and learns risk semantics via the proposed online error supervision. Second, uncertainty-gated fusion (UGF) is a lightweight gated fusion block that modulates language injection strength conditioned on uncertainty, strengthening cross-modal constraints in uncertain regions while suppressing ineffective perturbations in confident ones. Third, uncertainty-driven local refinement (UDLR) generates shape-adaptive soft refinement masks from uncertainty and guides a lightweight residual branch to focus corrections on high-risk regions (e.g., boundaries and thin structures), reducing merging/bleeding, discontinuities, and boundary shifts. Overall, without modifying the two-tower backbone, our method provides a unified mechanism for online risk learning during training and risk-guided fusion and correction during inference, matching the pronounced spatial nonuniformity of reliability in remote-sensing referring segmentation.

Our contributions are summarized as follows.

1) We propose a risk-aware framework for remote-sensing referring segmentation, introducing a pixel-wise cross-modal referential uncertainty map as a unified representation of spatial risk to guide inference.
2) We design RUS and an online error-aligned supervision strategy to make uncertainty prediction semantically consistent with model errors, yielding an interpretable spatial risk prior.
3) We propose two plug-and-play modules, UGF and UDLR, to adaptively control language modulation and to perform shape-adaptive local refinement driven by uncertainty, improving disambiguation and boundary/detail quality in high-risk regions while preserving stability in low-risk regions.

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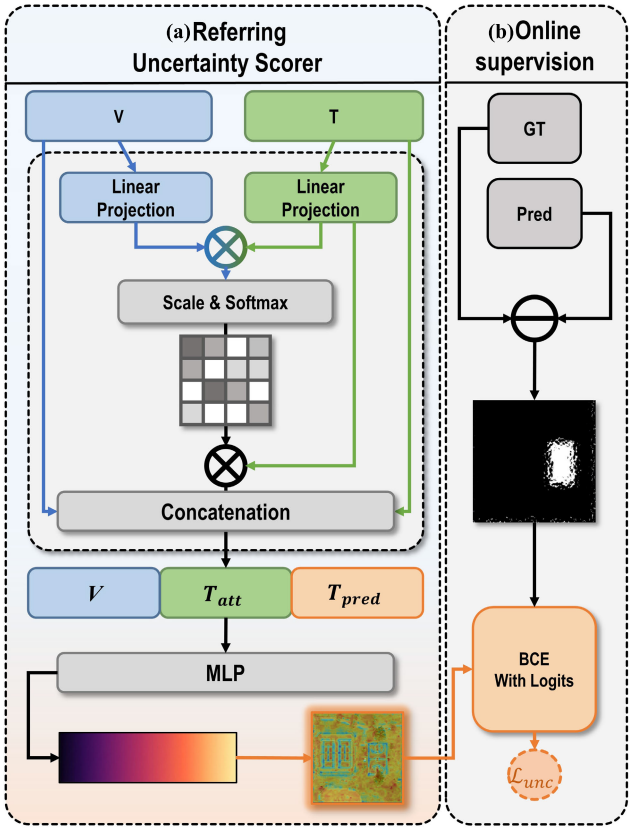


Fig. 2. Overview of RUS. (a) Fuses visual (V) and textual (T) features via cross-attention and raw broadcasting to predict uncertainty maps. (b) Trains with an error-consistency loss ($\mathcal{L}_{unc}$) derived from the prediction-GT discrepancy ($\ominus$).

4) We build a risk-guided fusion–local correction pipeline that enhances robustness and geometric fidelity in complex remote-sensing scenes without changing the backbone architecture.

II. Related Work

A. RIS in Natural Scenes

RIS requires fine-grained semantic alignment between visual and linguistic modalities. The field has evolved from CNN–RNN architectures to Transformer-based and generative paradigms.

1) Evolution of Architectures: Early methods adopted a "concat-and-process" pipeline. Hu et al. [11] pioneered the task using CNNs and LSTMs with simple concatenation. To enhance cross-modal interaction, attention mechanisms became standard. Shi et al. [12] focused on linguistic keywords, while Ye et al. [13] employed cross-modal self-attention. Recognizing the complexity of language, subsequent works explored progressive fusion strategies. Feng et al. [14] introduced an encoder fusion network to interleave features early, and Huang et al. [15] proposed a progressive comprehension framework to refine segmentation iteratively. Furthermore, multitask learning frameworks [16] have been explored to jointly optimize referring expression comprehension and segmentation.

2) Transformer and Generative Models: The introduction of Transformers [17] revolutionized RIS. VLT [18] and LAVT [5] leverage hierarchical vision Transformers to perform early vision–language fusion, suppressing background noise effectively. Recent trends include foundation model adaptation, such as CRIS [19] using CLIP [20], and generative approaches like SeqTR [21] and PolyFormer [22] that frame segmentation as sequence prediction. Additionally, efforts have been made to enhance robustness against adversarial or ambiguous inputs [23], and to model temporal–spatial context in dynamic scenes.

Despite these successes in object-centric natural images, direct transfer to remote sensing is hindered by the domain gap, specifically the lack of canonical viewpoints and the prevalence of dense, tiny objects.

B. Referring Remote Sensing Image Segmentation

RS-RefSeg extends grounding to overhead imagery, addressing challenges like extreme scale variation and rotation. The release of datasets such as GeoRef [24], RefSeg-RS [6], and the large-scale SkyEye [25] has established the benchmark for this field.

1) Scale and Rotation Invariance: Initial works focused on geometric adaptations. Yuan et al. [24] addressed scale disparity via global–local attention. Recent works continue to refine multiscale reasoning; for instance, Shi and Zhang [26] developed a multimodal multiscale fusion network to capture objects at varying resolutions, while Lin et al. [27] emphasized global–local reasoning to contextualize targets within broad geographic scenes.

2) Cross-Modal Alignment and Refinement: Distinguishing targets from similar distractors in dense clusters remains difficult. Huang et al. [15] proposed cross-modal progressive fusion to align features step by step. To improve boundary precision, Li et al. [28] designed a bilateral alignment network. Adapting large vision–language models (LVLMs) is also a growing trend, with works like RS-Prompter [29] and GeoRSCLIP [30] demonstrating strong generalization [31].

While existing methods excel at feature extraction and alignment, they typically apply a uniform inference mechanism across the image. They lack spatial risk awareness, treating ambiguous regions (e.g., shadows and crowded parking lots) with the same confidence as clear backgrounds. Our work postulates that explicitly modeling this uncertainty is key to breaking the current performance ceiling.

C. Cross-Modal Uncertainty Estimation

Uncertainty quantification [32] distinguishes between aleatoric (data) and epistemic (model) uncertainty [33].

1) Uncertainty in Vision: In standard segmentation, Bayesian approximations like MC Dropout [34] and Deep Ensembles [35] are classic but costly. Related Bayesian formulations have also been explored in cross-modal settings such as VQA question generation. Deterministic methods [36], [37] offer efficiency. Generative models, such as the probabilistic U-Net [38], model the distribution of plausible segmentations

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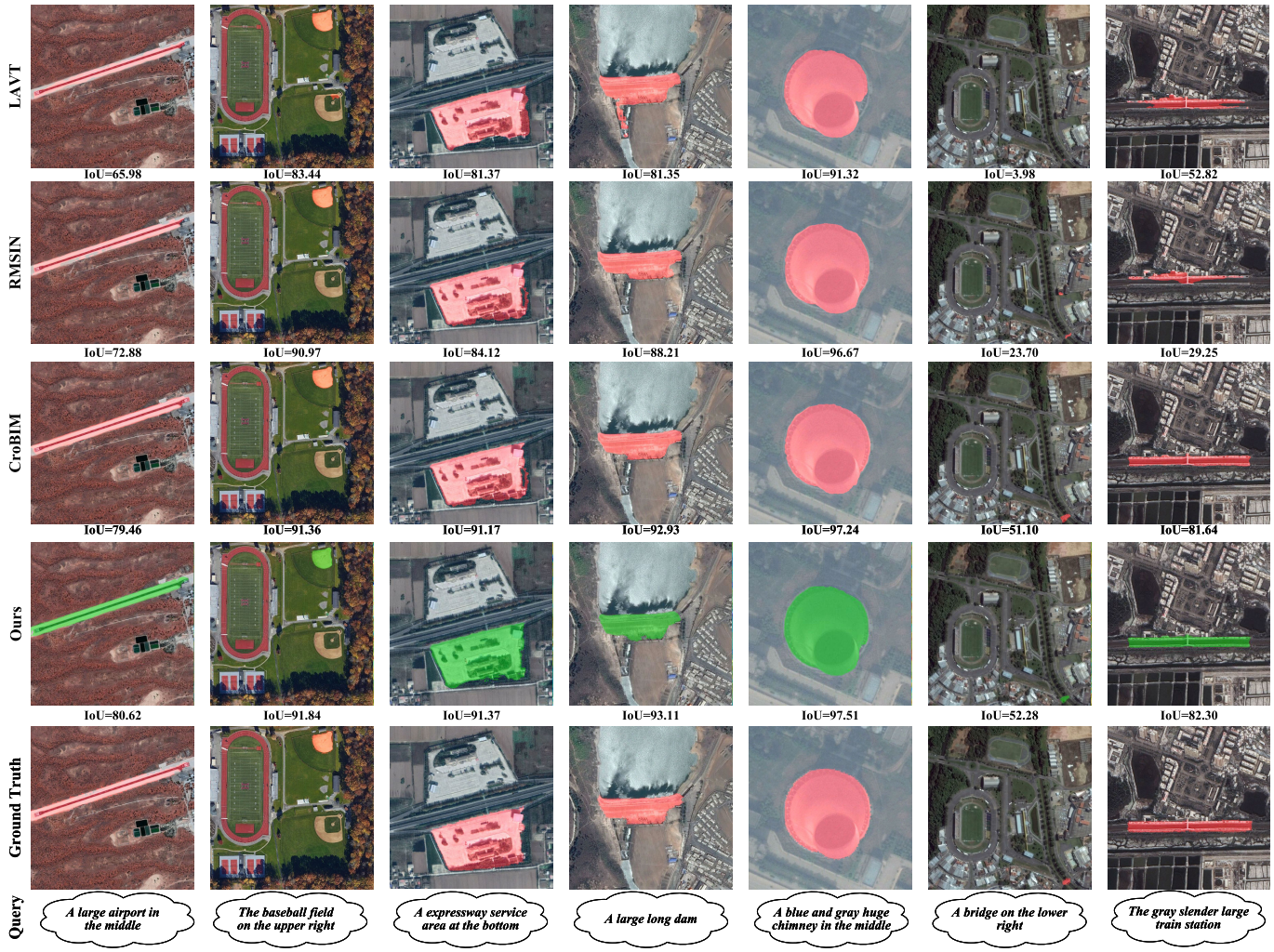


Fig. 3. Visualization of segmentation results for CroBIM and comparison methods on the RRSIS-D dataset test set, with corresponding IoU scores displayed.

for ambiguous images. These concepts have been applied to 3-D detection [39], zero-shot learning [40], and medical imaging [41], [42], providing general frameworks for reliability [43].

2) Uncertainty in Remote Sensing: In RS, uncertainty has primarily been used as a passive indicator for postprocessing in land-cover classification [44], change detection, or active learning [45]. Boundary-aware graph reasoning [46] and context-aware networks [47] implicitly handle ambiguity but do not explicitly quantify it.

Unlike prior works that use uncertainty merely for quality assessment, we propose an active uncertainty-guided framework. We leverage the cross-modal uncertainty map as a dynamic weight to orchestrate feature refinement, forcing the model to reattend to high-risk regions where visual–linguistic alignment is fragile.

III. Methodology

A. Overview

We study remote-sensing referring segmentation: given a remote-sensing image I and a text description S, the goal is to predict the pixel-wise mask of the referred target $\hat{Y} \in [0,1]^{H\times W}$ [1]. A central challenge is the spatially varying reliability of both cross-modal grounding and pixel prediction. On the one hand, abundant visually similar instances and large-scale variations in remote-sensing imagery induce ambiguities in vision–language alignment [2], [4]; on the other hand, thin or fragmented structures and blurred boundaries cause errors to cluster around object boundaries and fine details. As a result, uniformly injecting language cues or allocating uniform refinement across the entire image often leads to two coupled issues: unnecessary language-induced interference in reliable regions, insufficient disambiguation, and inadequate compute allocation in high-risk regions.

To explicitly capture such a spatial risk distribution, we introduce a pixel-level cross-modal grounding uncertainty map U as the key intermediate representation. Unless otherwise specified, U is represented in logit form; when a probabilistic uncertainty is needed, we use $U^p = \sigma(U) \in [0,1]$, where $\sigma(\cdot)$ denotes the Sigmoid function. By learning U with online supervision during training, the model predicts where errors are more likely to occur. At inference time, U serves as a feed-forward spatial prior for selective fusion and focused local

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Fig. 4. Visualization of segmentation results for CroBIM and comparison methods on the RefSegRS dataset, with corresponding IoU scores displayed.

refinement, forming a risk-aware pipeline tailored to remote-sensing referring segmentation.

Most existing methods adopt a two-tower architecture [5], [18]: a visual encoder and a text encoder extract representations separately, followed by a pixel-level decoder that performs cross-modal fusion and produces the segmentation output. Concretely, the visual encoder extracts multiscale visual features while the text encoder maps S to token embeddings. During progressive decoding, cross-modal interactions produce a foreground logit map $P_{\text{fg}} \in \mathbb{R}^{B\times 1\times H\times W}$ and an initial mask prediction $\hat{Y} = \sigma(P_{\text{fg}})$.

Built upon this backbone, we introduce three plug-and-play modules that share a unified intermediate representation—the grounding uncertainty map U.

1) RUS is placed between the encoder and the decoder. It takes visual and textual tokens as inputs and predicts U to quantify spatial grounding ambiguity and local error risk.
2) UGF is applied once as a postfusion step before decoding: it modulates the strength of language injection according to U, so that textual constraints primarily affect high-uncertainty locations while suppressing interference in low-uncertainty regions.
3) UDLR derives shape-adaptive soft regions from U^p to guide a refinement branch, concentrating corrections on error-prone areas such as boundaries and fine structures.

Together, these components instantiate a training-time "online risk learning" loop (via RUS supervision) and an inference-time "risk-guided fusion–local correction" pipeline in Fig. 1.

B. Backbone and Interaction Stages

Following two-tower referring segmentation pipelines [5], we instantiate the framework with a ConvNeXt-B visual encoder [48], a BERT text encoder [49], and a Mask2Former-style multiscale mask decoder [50]. The visual encoder outputs a four-level feature pyramid at strides $\{4, 8, 16, 32\}$ with channel dimensions $\{256, 512, 1024, 2048\}$. Each level is projected to a unified embedding dimension ($C = 256$) and flattened into visual tokens $\mathbf{V}^{(s)} \in \mathbb{R}^{B\times N_s\times C}$ with $N_s = H_s W_s$. The text encoder produces token embeddings, which are projected to the same dimension, yielding $\mathbf{T} \in \mathbb{R}^{B\times L\times C}$ with a padding

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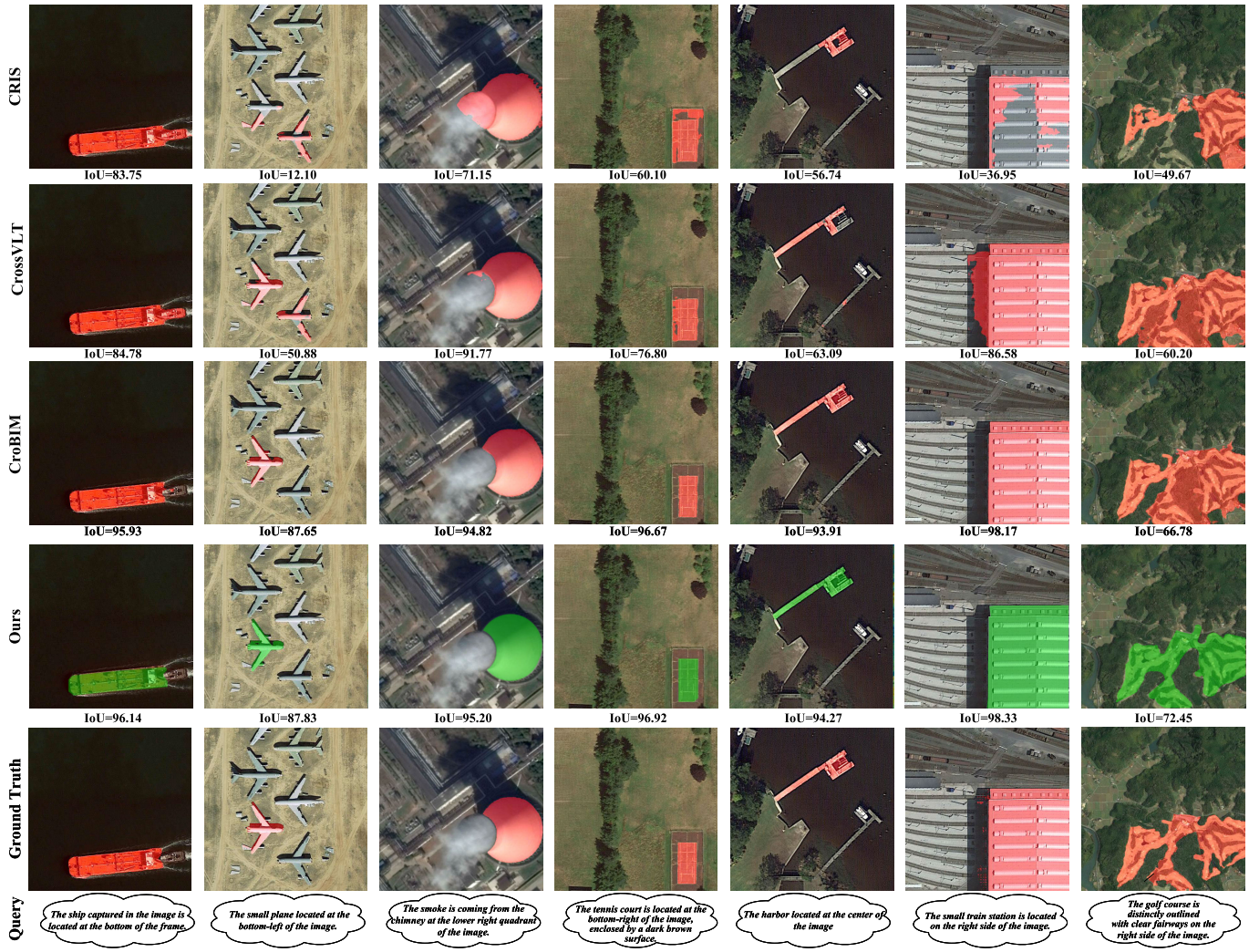


Fig. 5. Visualization of segmentation results for CroBIM and comparison methods on the RisBench dataset, with corresponding IoU scores displayed.

mask $\mathbf{m} \in \{0,1\}^{B\times L}$ (where $\mathbf{m} = 1$ indicates the token is ignored).

The decoder follows a Mask2Former-style design [50] with Transformer-based multiscale interaction. In our instantiation, the multiscale Transformer encoder contains six layers with eight attention heads and operates on three feature levels (strides $\{8, 16, 32\}$). Let $\Phi^{(s)}(\cdot)$ denote one interaction block that updates visual tokens by cross-modal interaction and subsequent token mixing. At stage s, we have

$$\begin{aligned}\mathbf{V}^{(s)+} &= \Phi^{(s)}\left(\mathbf{V}^{(s)}, \mathbf{T}, \mathbf{m}\right)\\ \mathbf{F}^{(s)+} &= \text{reshape}\left(\mathbf{V}^{(s)+}\right)\end{aligned} \tag{1}$$

where reshape$(\cdot)$ restores the spatial layout (H_s, W_s). The decoder finally outputs foreground logits $P_{\text{fg}} \in \mathbb{R}^{B\times 1\times H\times W}$, producing $\hat{Y} = \sigma(P_{\text{fg}})$.

RUS predicts an uncertainty logit map $U \in \mathbb{R}^{B\times 1\times H_u\times W_u}$ at token resolution. In our design, RUS operates on the coarsest (stride-32) visual tokens, which have the largest receptive field; thus $(H_u, W_u) = (H_{32}, W_{32})$. When used at a decoder stage s or at full resolution, we align U by bilinear interpolation on logits

$$\begin{aligned}U^{(s)} &= \text{Resize}\left(U, H_s, W_s\right)\\ \tilde{U} &= \text{Resize}\left(U, H, W\right).\end{aligned} \tag{2}$$

Interpolating in logit space avoids probability saturation and provides a consistent spatial prior for UGF and local refinement.

C. Referring Uncertainty Scorer

Errors in remote-sensing referring segmentation are often spatially clustered: semantically similar land-cover types induce confusion, while thin/fractured structures and blurred boundaries cause locally unstable predictions [1]. To explicitly model this location-dependent risk distribution before decoding and multiscale fusion, we insert an RUS between the encoder and the decoder to predict a pixel-wise referring uncertainty map U in Fig. 2.

1) Architecture: Let $\mathbf{V} \in \mathbb{R}^{B\times N_u\times C}$ denote the stride-32 visual tokens (after projection to $C = 256$), and $\mathbf{T} \in \mathbb{R}^{B\times L\times C}$ denote the projected text tokens. RUS outputs an uncertainty

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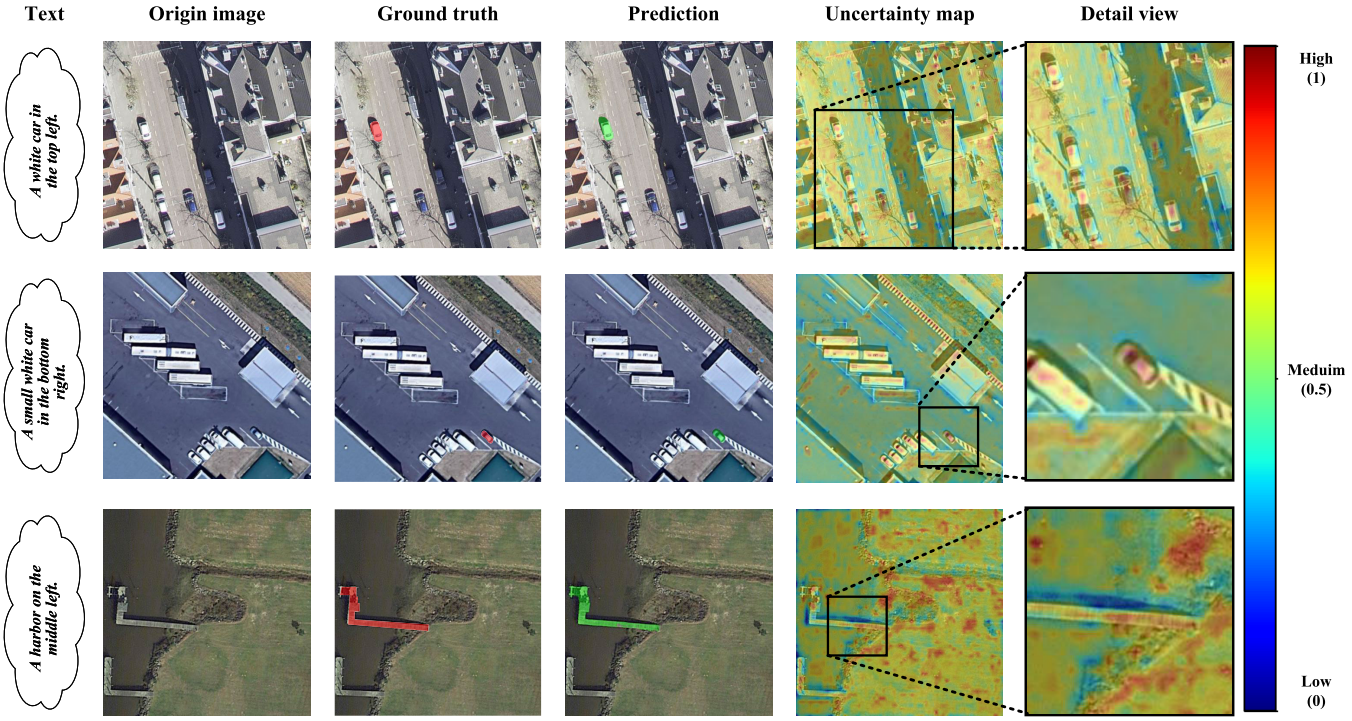


Fig. 6. Qualitative visualization of the uncertainty prior predicted by RUS. From left to right: the referring expression, original image, ground-truth mask (red), model prediction (green), the predicted uncertainty probability map $U^p \in [0, 1]$ (normalized; warmer colors indicate higher uncertainty, see colorbar), and a zoomed-in detail view of the highlighted region. The uncertainty responses concentrate on hard regions such as small/low-contrast targets, confusable surroundings, and boundary/thin-structure areas, indicating that RUS captures an error-prone spatial risk prior for subsequent uncertainty-guided fusion and local refinement.

logit $\mathbf{u} \in \mathbb{R}^{B\times N_u\times 1}$ for each visual position, and reshapes it into $U \in \mathbb{R}^{B\times 1\times H_u\times W_u}$. We define U as an indicator of text-referential matching ambiguity/local error risk: a larger value suggests that the position is more likely to suffer from confusion among similar objects or boundary/detail mis-segmentation, and subsequent modules should enforce stronger textual constraints or allocate more refinement capacity there.

To extract position-level referential ambiguity cues with minimal overhead, we apply linear projections to the visual and textual features (initialized as identity/near-identity after projection for stable early training) and employ scaled dot-product attention to perform "visual-position to text-token" alignment and aggregation. We then concatenate the original visual token with the aligned textual feature and feed them into a lightweight MLP to output the uncertainty logit

$$\begin{aligned}
\mathbf{V}' &= \mathbf{V}\mathbf{W}_v, \quad \mathbf{T}' = \mathbf{T}\mathbf{W}_t \\
\mathbf{A} &= \mathrm{Softmax}\left(\gamma \cdot \mathbf{V}'\mathbf{T}'^{\top}\right) \\
\tilde{\mathbf{T}} &= \mathbf{A}\mathbf{T}' \\
\mathbf{u} &= \mathrm{MLP}\left(\left[\mathbf{V}; \tilde{\mathbf{T}}\right]\right).
\end{aligned} \tag{3}$$

Here, Softmax$(\cdot)$ is applied along the text-sequence dimension, γ is a learnable scaling factor initialized to the standard attention scale (e.g., $1/\sqrt{C}$), and $\mathbf{W}_v, \mathbf{W}_t$ are linear projection matrices. Finally, $U = \mathrm{reshape}(\mathbf{u})$ recovers the spatial layout (H_u, W_u). The RUS output is kept in logit form (without Sigmoid), matching the subsequent gating mechanism and the training-time supervision for numerical consistency.

2) Online Supervision for Risk Semantics: Architecture alone does not guarantee that U learns the desired "risk hint" semantics; therefore, we construct an online pixel-wise supervision signal during training. Let the decoder output foreground logits $P_{\mathrm{fg}} \in \mathbb{R}^{B\times 1\times H\times W}$. We obtain a binary prediction

$$\hat{y} = \mathrm{StopGrad}\left(\mathbb{I}\left(\sigma\left(P_{\mathrm{fg}}\right) > 0.5\right)\right) \tag{4}$$

and build a pixel-wise error map

$$e(x) = \mathbb{I}\left(\hat{y}(x) \neq y(x)\right) \tag{5}$$

where $y(x)$ is the GT and StopGrad$(\cdot)$ prevents gradients from backpropagating through the online target.

The online error map is treated as a dynamic pseudo-target for the uncertainty branch. If gradients from $\mathcal{L}_{\mathrm{unc}}$ were allowed to backpropagate into the segmentation backbone, it could create a degenerate incentive: the backbone might alter its predictions/features to make errors easier to predict (i.e., "self-confirming shortcuts"), rather than improving segmentation quality. Therefore, we adopt a *forward coupling, backward decoupling* strategy via StopGrad$(\cdot)$, ensuring that $\mathcal{L}_{\mathrm{seg}}$ governs semantic learning while $\mathcal{L}_{\mathrm{unc}}$ focuses purely on spatial reliability estimation.

We fix the supervision target as $z(x) = e(x)$, so that a larger U consistently corresponds to higher error risk. Optionally, we apply a light local smoothing (e.g., small-kernel blur) to suppress isolated noisy pixels, yielding $z(x) = \mathrm{Blur}(e(x))$.

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Since the resolution of U is lower than that of the final segmentation, we align it to the segmentation resolution to obtain $\tilde{U} \in \mathbb{R}^{B\times 1\times H\times W}$ and optimize a weighted per-pixel BCEWithLogits loss

$$\mathcal{L}_{\text{unc}} = \frac{1}{BHW}\sum_{b,x} w_b^{\text{sample}} \cdot w_{b,x}^{\text{pixel}} \cdot \text{BCEWithLogits}\left(\tilde{U}_b(x), z_b(x)\right). \tag{6}$$

We use two lightweight reweighting terms for stable online risk learning.

1) Sample-level difficulty weight

$$w_b^{\text{sample}} = 1 + \lambda_s \cdot \mathbb{I}\left(\text{IoU}\left(\hat{y}_b, y_b\right) < \delta\right) \tag{7}$$

which upweights globally hard samples via a fixed difficulty threshold δ in the RUS sample reweighting term of (7) (default $\delta = 0.5$ and $\lambda_s = 1$ unless otherwise stated).

2) Pixel-level class rebalancing weight to counter the sparsity of error pixels

$$w_{b,x}^{\text{pixel}} = \begin{cases} \dfrac{n_b^0 + \epsilon}{n_b^1 + \epsilon}, & \text{if } z_b(x) = 1 \\ 1, & \text{if } z_b(x) = 0 \end{cases} \tag{8}$$

where n_b^1 and n_b^0 denote the counts of positive/negative pixels in z_b and ϵ is a small constant.

To mitigate early-stage noise in the online supervision, we enable $\mathcal{L}_{\text{unc}}$ only after the first $T_{\text{warm}} = 0.1E$ training epochs, where E is the total number of epochs. For spatial smoothing, we use a Gaussian blur for $z(x) = \text{Blur}(e(x))$ with kernel size $k_{\text{blur}} = 7$ and $\sigma = 1.0$.

Finally, $\mathcal{L}_{\text{unc}}$ complements the segmentation loss $\mathcal{L}_{\text{seg}}$: $\mathcal{L}_{\text{seg}}$ learns what to segment, while $\mathcal{L}_{\text{unc}}$ learns where referring/segmentation errors are more likely. Through online supervision, RUS explicitly encodes this distribution into U, providing consistent spatial guidance for UGF and soft-region local refinement.

D. Uncertainty-Gated Fusion

In remote-sensing referring segmentation, linguistic cues exhibit a dual effect of disambiguation and perturbation. In regions crowded with visually similar objects, textual constraints are crucial for distinguishing the target; however, uniformly injecting language information even where visual evidence is already sufficient can induce referring drift and amplify mis-segmentation [4]. To address this spatially nonuniform reliability, we propose UGF, which uses the pixel-wise referring uncertainty predicted by RUS to modulate cross-modal updates at each position, enabling on-demand language allocation: strengthening disambiguation in high-risk regions while suppressing ineffective perturbations in low-risk regions.

UGF is instantiated as a single postfusion module applied between the encoder and the decoder (i.e., not repeated across decoding stages). Let $\mathbf{V} \in \mathbb{R}^{B\times N\times C}$ denote visual tokens at a given feature level involved in postfusion (in practice, we apply the same UGF to each of the three decoder input levels with shared parameters) and let $\mathbf{T} \in \mathbb{R}^{B\times L\times C}$ be the text tokens with padding mask $\mathbf{m} \in \{0,1\}^{B\times L}$. UGF computes cross-attention with visual tokens as queries and text tokens as keys/values and obtains a visual increment via a projection

$$\Delta\mathbf{V} = \text{Proj}\left(\text{CrossAttn}\left(\mathbf{V}, \mathbf{T}, \mathbf{m}\right)\right) \tag{9}$$

where CrossAttn$(\cdot)$ is multihead attention and Proj$(\cdot)$ corresponds to Linear+Dropout in implementation to stabilize the update scale.

Since RUS predicts uncertainty logits U at stride-32 resolution, we align them to the token resolution of the fused level by bilinear resizing on logits and flattening

$$\begin{aligned} U^* &= \text{Resize}\left(U, H_*, W_*\right) \\ \mathbf{u} &= \text{Flatten}\left(U^*\right) \in \mathbb{R}^{B\times N_*\times 1} \end{aligned} \tag{10}$$

where $N_* = H_* W_*$. We map uncertainty to fusion coefficients using a Sigmoid with a learnable affine transform

$$\mathbf{g} = \sigma\left(\alpha\mathbf{u} + \beta\right) \tag{11}$$

where α and β are learnable scalars (initialized to encourage a monotonic mapping so that higher U yields stronger fusion). Finally, UGF injects the gated update in a residual manner and applies layer normalization

$$\mathbf{V}^+ = \text{LN}\left(\mathbf{V} + \mathbf{g} \odot \Delta\mathbf{V}\right). \tag{12}$$

This design explicitly converts uncertainty cues into fusion-strength control: positions with high uncertainty receive stronger textual constraints for disambiguation, whereas positions with low uncertainty suppress cross-modal perturbations to prevent the spread of linguistic noise. As an independent postfusion module, UGF improves robustness under complex descriptions and highly confusable scenes without altering the backbone or the decoder interfaces.

E. Uncertainty-Driven Local Refinement

Errors in remote-sensing oriented segmentation are typically concentrated around boundaries and fine-grained structures: targets are often slender, fragmented, or weak-contrast, and lie adjacent to background objects (e.g., along roads, shorelines, and building edges) [1], [7]. As a result, coarse predictions from the main decoder are prone to local adhesion, breakage, and boundary misalignment. To address these high-risk, shape-irregular local errors, we propose UDLR: the uncertainty map determines where secondary corrections are needed, and a continuous soft region replaces Top-K square ROIs (as used in PointRend [51]), thereby reducing boundary artifacts and better matching the true geometry of remote-sensing targets.

Let the high-resolution mask feature be $\mathbf{F}$ (the highest-resolution decoder feature), the coarse foreground logit be P_{fg}, and the uncertainty probability map be $U^p = \sigma(\tilde{U}) \in [0,1]$ (obtained by applying Sigmoid to the aligned uncertainty logit $\tilde{U}$). UDLR first maps U^p to a continuous soft refinement mask $\mathbf{M}$, which jointly represents the refinement region and its strength. We construct $\mathbf{M}$ via a temperature-scaled thresholded Sigmoid

$$\mathbf{M} = \sigma\left(\frac{\text{Blur}\left(U^p\right) - \tau}{t}\right) \tag{13}$$

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where τ is the fixed central threshold that determines the refinement region in (13) and t is the temperature that controls the smoothness of the transition (default $\tau = 0.5$ and $t = 0.1$ unless otherwise stated). Here, Blur$(\cdot)$ denotes an optional local Gaussian smoothing operator applied to U^p (kernel size $k_{\text{blur}} = 7$ and $\sigma = 1.0$ by default). To improve the stability of region selection, $\mathbf{M}$ is used only as a selection/weighting signal, and gradients are stopped through $\mathbf{M}$

$$\mathbf{M} \leftarrow \text{StopGrad}(\mathbf{M}). \tag{14}$$

Next, a lightweight convolutional refinement head $\mathcal{R}(\cdot)$ predicts a logit residual Δ at full resolution from the high-resolution feature, the coarse prediction, and uncertainty

$$\Delta = \mathcal{R}\left(\left[\mathbf{F}; P_{\text{fg}}; U^p\right]\right) \tag{15}$$

where $\mathcal{R}(\cdot)$ consists of two 3×3 convolution layers with ReLU activations, followed by a 1×1 convolution to output Δ. To start from near-identity refinement and improve stability, the final 1×1 projection is initialized to produce near-zero residual. Finally, UDLR spatially weights the residual with the soft mask, correcting the coarse prediction only in high-uncertainty regions

$$P_{\text{fg}}^{\text{ref}} = P_{\text{fg}} + \mathbf{M} \odot \Delta. \tag{16}$$

Optionally, we apply auxiliary supervision on the refined prediction

$$\mathcal{L}_{\text{ref}} = \mathcal{L}_{\text{seg}}\left(\sigma\left(P_{\text{fg}}^{\text{ref}}\right), y\right) \tag{17}$$

and the overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_{\text{unc}}\mathcal{L}_{\text{unc}} + \lambda_{\text{ref}}\mathcal{L}_{\text{ref}} \tag{18}$$

where λ_{ref} is set to 0 when the auxiliary refinement supervision is disabled.

This design offers two key advantages: updates are explicitly confined to high-response regions of $\mathbf{M}$, reducing unnecessary perturbations in low-uncertainty areas; and refinement regions are adaptively generated from the spatial structure of U^p, naturally conforming to slender or irregular boundaries and avoiding hard-edge artifacts of fixed ROIs. By sharing the same prior $\tilde{U}$ (logit)/U^p (probability) with uncertainty modeling, UDLR aligns refinement with the model's risk assessment, thereby consistently improving segmentation quality around boundaries and fine details.

F. Computation Overhead

The proposed modules introduce lightweight overhead on top of the two-tower backbone. RUS consists of one cross-modal alignment step and a shallow MLP for token-wise uncertainty prediction at the coarsest feature level. UGF adds a single postfusion cross-attention update with a simple scalar-gated modulation (applied once before decoding), and therefore minimally increases computation. UDLR employs a shallow convolutional residual predictor and applies corrections through an uncertainty-derived soft mask, focusing effective updates on uncertain regions. Overall, the method remains practical for remote-sensing referring segmentation while improving robustness under ambiguous referring conditions and fine-structure errors.

IV. Experiments

A. Datasets and Evaluation Metrics

1) Datasets: To comprehensively evaluate the effectiveness and robustness of the proposed framework, we conduct experiments on three challenging remote sensing referring segmentation benchmarks: **RefSegRS** [1], **RRSIS-D** [1], and **RISBench** [2]. These datasets are selected to cover a wide range of difficulties, including high-resolution targets, complex semantic attributes, and large-scale resolution variations. The detailed statistics and data partition for each dataset are summarized in Table I.

1) *RefSegRS [1]:* As a high-resolution benchmark, RefSegRS is characterized by its fine-grained imagery (0.13-m spatial resolution). It poses significant challenges for the precise boundary delineation of objects in complex scenes.
2) *RRSIS-D [1]:* This dataset distinguishes itself with rich semantic annotations. Unlike standard datasets, it includes 20 object categories and seven specific attributes, requiring the model to capture intricate linguistic cues and geometric details (e.g., orientation).
3) *RISBench [2]:* Being a large-scale dataset, RISBench introduces extreme scale variations with spatial resolutions spanning from 0.1 to 30 m. This diversity tests the model's capability to handle objects with significant size differences across varying observation altitudes.

2) Evaluation Metrics: Following the standard evaluation protocols in referring segmentation [1], [5], we adopt overall intersection-over-union (oIoU), mean intersection-over-union (mIoU), and precision at thresholds Pr@X ($X \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$) as our primary metrics.

oIoU focuses on the overall pixel-level accuracy and is calculated by accumulating the intersection and union areas over the entire dataset. It gives more weight to large-scale objects

$$\text{oIoU} = \frac{\sum_{i=1}^{N} \mathcal{I}_i}{\sum_{i=1}^{N} \mathcal{U}_i} \tag{19}$$

where N is the total number of test samples and $\mathcal{I}_i$ and $\mathcal{U}_i$ represent the intersection and union areas between the predicted mask and the GT for the ith sample, respectively.

mIoU provides a macroaverage performance by treating each sample equally, regardless of the object size

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \frac{\mathcal{I}_i}{\mathcal{U}_i}. \tag{20}$$

Additionally, **Pr@X** measures the percentage of test samples where the IoU score exceeds the threshold X, reflecting the reliability of the model under different strictness levels.

3) Implementation Details: For reproducibility, we summarize key hyperparameters in Table II. We enable $\mathcal{L}_{\text{unc}}$ after the first $T_{\text{warm}} = 0.1E$ epochs (E is total epochs). For the error-map smoothing, we use a Gaussian blur with kernel size $k_{\text{blur}} = 7$ and $\sigma = 1.0$. In UGF, the gating parameters are initialized as $\alpha = 1.0$ and $\beta = 0.0$. In UDLR, we use $\tau = 0.5$ and temperature $t = 0.1$ by default. Unless specified, we set

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TABLE I
STATISTICAL DETAILS OF THE THREE DATASETS USED IN OUR EXPERIMENTS

Dataset	Resolution	Image Size	Split (Image-Text Pairs)		
			Train	Val	Test
RefSegRS	0.13 m	512 × 512	2,172	431	1,817
RRSIS-D	-	800 × 800	12,181	1,740	3,481
RISBench	0.1 m - 30 m	512 × 512	26,300	10,013	16,159

TABLE II
CONFIGURATION OF CRITICAL HYPERPARAMETERS. E DENOTES THE TOTAL TRAINING EPOCHS

Symbol	Parameter	Value
T_{warm}	Warm-up epochs	$0.1E$
k	Neighbors	7
k_{blur}	Gaussian kernel size	7
σ	Gaussian σ	1.0
δ	Difficulty threshold (Eq. 7)	0.5
λ_{unc}	Loss weight for $\mathcal{L}_{\text{unc}}$ (Eq. 18)	0.1
λ_{ref}	Loss weight for $\mathcal{L}_{\text{ref}}$ (Eq. 18)	0.1
α	UGF gate init	1.0
β	UGF gate init	0.0
τ	UDLR threshold (Eq. 13)	0.5
t	UDLR temperature (Eq. 13)	0.1

$\delta = 0.5$ in (7), $\lambda_{\text{unc}} = 0.1$ in (18), and $\lambda_{\text{ref}} = 0.1$ when auxiliary refinement supervision is enabled (otherwise $\lambda_{\text{ref}} = 0$).

B. Experimental Setup

1) Network Configuration: Our framework is implemented on the PyTorch platform [52]. To ensure comprehensive feature extraction, we employ ConvNeXt (ConvNeXt-Base) [48]. Specifically, ConvNeXt-Base utilizes weights derived from the self-supervised SMLFR algorithm (generative ConvNet foundation model) [10], which is designed to capture sparse and low-frequency structures in remote sensing images. For linguistic processing, we adopt the standard 12-layer BERT-base model [49] with a hidden dimension of 768. The input images are uniformly resized to 480 × 480 pixels, and the maximum length for textual descriptions is truncated to 20 tokens. We adopt a standard BERT fixed-length protocol (padding + attention mask) with `max_tokens` = 20. A statistical analysis on the RRSIS-D training set shows that the median text length is eight tokens and the 95th percentile is 16 tokens; thus, truncation is triggered in only 0.14% of samples (17/12,181), with an average loss of fewer than 1.5 trailing tokens.

2) Training Protocol: The model is trained on a cluster of eight NVIDIA A800 GPUs with a total batch size of 32. We utilize the AdamW optimizer [53] with an initial learning rate of 5×10^{-5} and a weight decay coefficient of 0.01. The learning rate follows a polynomial decay schedule. The entire training process spans 40 epochs. Consistent with prior works, no additional data augmentation or postprocessing techniques are applied during the training and inference phases.

C. Comparison With State-of-the-Art Methods

1) RRSIS-D: To evaluate the effectiveness of our proposed method, we conducted extensive experiments on the RRSIS-D dataset, and the comparison results are reported in Table III. RRSIS-D provides rich semantic annotations with 20 object categories and seven specific attributes, where referring expressions frequently involve fine-grained linguistic cues and geometric details (e.g., orientation and spatial relations) [1]. Such characteristics require the model to align language with visual evidence more precisely, rather than relying solely on coarse category recognition.

We compared our framework with a wide range of methods, including classical LSTM-based models (e.g., RRN [54] and CMPC [15]) and recent CLIP/BERT-based approaches (e.g., CRIS [19], LAVT [5], RIS-DMMI [55], robust-ref-seg [23], and CARIS [56]). Overall, LSTM-based methods generally underperform compared to CLIP- and BERT-based solutions. This is mainly because RRSIS-D descriptions often contain attribute-level and relation-level constraints, for which capturing global semantic dependencies is crucial. Consequently, modern combinations of hierarchical visual encoders (e.g., Swin Transformer [57] or ConvNeXt [48]) and BERT-style text encoders constitute strong baselines on this dataset.

Among all compared methods, our framework achieves superior performance. The baseline CroBIM (with ConvNeXt-B) already delivers competitive results and surpasses the previous state-of-the-art method CARIS [56] in terms of mIoU. More importantly, CroBIM-U further improves performance by explicitly addressing referential uncertainty, which is particularly relevant to RRSIS-D, where multiple instances may share similar categories but differ in attributes (e.g., "the rotated/leftmost/elongated object"). By reducing ambiguity in language-to-region matching, CroBIM-U achieves the best results across almost all metrics.

Specifically, CroBIM-U achieves a Pr@0.5 of 76.31% on the validation set and 75.60% on the test set, outperforming CARIS (71.61%/71.50%) by over 4%. In terms of mIoU, which better reflects segmentation quality across diverse categories and attribute-conditioned targets, CroBIM-U reaches 66.07% (Val) and 65.07% (Test), bringing substantial gains of +3.19% and +2.95% over CARIS, respectively.

Although robust-ref-seg [23] reports a slightly higher oIoU on the test set, CroBIM-U achieves the highest oIoU on the validation split (77.83%). The combination of higher mIoU and competitive oIoU suggests that our method is particularly effective in improving the segmentation of attribute-sensitive and potentially smaller or visually similar targets, rather than being dominated by large-area classes. These results validate the effectiveness of our design in achieving precise pixel-level localization under rich semantic and geometric constraints on RRSIS-D in Fig. 3

2) RefSegRS: We conduct experiments on the RefSegRS dataset, and the results on both the validation and test sets are reported in Table IV. RefSegRS is a high-resolution benchmark with fine-grained imagery (0.13-m spatial resolution), where objects often exhibit subtle boundaries and appear in cluttered scenes [1]. Therefore, metrics under stricter overlap

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TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RRSIS-D DATASET. OPTIMAL AND SUBOPTIMAL PERFORMANCE IN EACH METRIC ARE MARKED BY RED AND BLUE, RESPECTIVELY

Method	Visual Encoder	Text Encoder	Pr@0.5		Pr@0.6		Pr@0.7		Pr@0.8		Pr@0.9		oIoU		mIoU	
			Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
RRN [54]	ResNet-101	LSTM	51.09	51.07	42.47	42.11	33.04	32.77	20.80	21.57	6.14	6.37	66.53	66.43	46.06	45.64
CSMA [13]	ResNet-101	None	55.68	55.32	48.04	46.45	38.27	37.43	26.55	25.39	9.02	8.15	69.68	69.43	48.85	48.54
LSCM [58]	ResNet-101	LSTM	57.12	56.02	48.04	46.25	37.87	37.70	26.35	25.28	7.93	7.86	69.28	69.10	50.36	49.92
CMPC [15]	ResNet-101	LSTM	57.93	55.83	48.85	47.40	36.94	35.28	25.25	25.45	9.31	9.20	70.15	69.41	51.01	49.24
BRINet [59]	ResNet-101	LSTM	58.79	56.90	49.54	48.77	39.65	38.61	28.21	27.03	9.19	8.93	70.73	69.68	51.41	49.45
CMPC+ [60]	ResNet-101	LSTM	59.19	57.95	49.41	48.31	38.67	37.61	25.91	24.33	8.16	7.94	70.80	70.13	51.63	50.12
BKINet [61]	ResNet-101	CLIP	58.79	56.90	49.54	48.77	39.65	39.12	28.21	27.03	9.19	9.16	70.78	69.89	51.14	49.65
ETRIS [62]	ResNet-101	CLIP	62.10	61.07	53.73	50.99	43.12	40.94	30.79	29.30	12.90	11.43	72.75	71.06	55.21	54.21
CRIS [19]	ResNet-101	CLIP	56.44	54.84	47.87	46.77	39.77	38.06	29.31	28.15	11.84	11.52	70.98	70.46	50.75	49.69
LGCE [1]	Swin-B	BERT	68.10	67.65	60.61	61.53	51.45	51.42	42.34	39.62	23.85	22.94	76.68	76.33	60.16	59.37
LAVT [5]	Swin-B	BERT	65.23	63.98	58.79	57.57	50.29	49.30	40.11	38.06	23.05	22.29	76.27	76.16	57.72	56.82
RMSIN [4]	Swin-B	BERT	68.39	67.16	61.72	60.36	52.24	50.16	41.44	38.72	23.16	22.81	**77.53**	75.79	60.23	58.79
CrossVLT [63]	Swin-B	BERT	67.07	66.42	59.54	59.41	50.80	49.76	40.57	38.67	23.51	23.30	76.25	75.48	59.78	58.48
RIS-DMMI [55]	Swin-B	BERT	70.40	68.74	63.05	60.96	54.14	50.33	41.95	38.38	23.85	21.63	77.01	76.20	61.70	60.25
robust-ref-seg [23]	Swin-B	BERT	64.22	66.59	58.72	59.58	50.00	49.93	35.78	38.72	24.31	23.30	76.39	**77.40**	58.92	58.91
CARIS [56]	Swin-B	BERT	71.61	71.50	64.66	63.52	54.14	52.92	42.76	40.94	23.79	**23.90**	77.48	**77.17**	62.88	62.12
CroBIM	Swin-B	BERT	74.20	**75.00**	66.15	66.32	54.08	54.31	41.38	41.09	22.30	21.78	76.24	76.37	63.99	64.24
CroBIM	ConvNeXt-B	BERT	**74.94**	74.58	**67.64**	**67.57**	**57.18**	**55.59**	**44.66**	**41.63**	**24.60**	23.56	76.94	75.99	**65.05**	**64.46**
CroBIM-U	ConvNeXt-B	BERT	**76.31**	**75.60**	**68.30**	**67.68**	**57.67**	**56.47**	**44.97**	**42.57**	**24.89**	**24.16**	**77.83**	76.70	**66.07**	**65.07**

TABLE IV

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE REFSEGRS DATASET. OPTIMAL AND SUBOPTIMAL PERFORMANCE IN EACH METRIC ARE MARKED BY RED AND BLUE, RESPECTIVELY

Method	Visual Encoder	Text Encoder	Pr@0.5		Pr@0.6		Pr@0.7		Pr@0.8		Pr@0.9		oIoU		mIoU	
			Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
RRN [54]	ResNet-101	LSTM	55.43	30.26	42.98	23.01	23.11	14.87	13.72	7.17	2.64	0.98	69.24	65.06	50.81	41.88
CMSA [13]	ResNet-101	None	39.24	26.14	38.44	18.52	20.39	10.66	11.79	4.71	1.52	0.69	63.84	62.11	43.62	38.72
LSCM [58]	ResNet-101	LSTM	56.82	31.54	41.24	20.41	21.85	9.51	12.11	5.29	2.51	0.84	62.82	61.27	40.59	35.54
BRINet [59]	ResNet-101	LSTM	36.86	20.72	35.53	14.26	19.93	9.87	10.66	2.98	2.84	1.14	61.59	58.22	38.73	31.51
MAttNet [64]	ResNet-101	LSTM	48.56	28.79	40.26	22.51	20.59	11.32	12.98	3.62	2.02	0.79	66.84	64.28	41.73	33.42
BKINet [61]	ResNet-101	CLIP	52.04	36.12	35.31	20.62	18.35	15.22	12.78	6.26	1.23	1.33	75.37	63.37	56.12	40.41
ETRIS [62]	ResNet-101	CLIP	54.99	35.77	35.03	23.00	25.06	13.98	12.53	6.44	1.62	1.10	72.89	65.96	54.03	43.11
CRIS [19]	ResNet-101	CLIP	53.13	35.77	36.19	24.11	24.36	14.36	11.83	6.38	2.55	1.21	72.14	65.87	53.74	43.26
RMSIN [4]	Swin-B	BERT	68.21	42.32	46.64	25.87	24.13	14.20	13.69	6.77	3.25	1.27	74.40	68.31	54.24	42.63
CrossVLT [63]	Swin-B	BERT	67.52	41.94	43.85	25.43	25.99	15.19	14.62	3.71	1.87	1.76	76.12	69.73	55.27	42.81
RIS-DMMI [55]	Swin-B	BERT	86.17	63.89	74.71	44.30	38.05	19.81	18.10	6.49	3.25	1.00	74.02	68.58	65.72	52.15
CARIS [56]	Swin-B	BERT	68.45	45.40	47.10	27.19	25.52	15.08	14.62	7.87	3.71	1.98	75.79	69.74	54.30	42.66
robust-ref-seg [23]	Swin-B	BERT	81.67	50.25	52.44	28.01	30.86	17.83	17.17	9.19	5.80	2.48	77.74	71.13	60.44	47.12
LGCE [1]	Swin-B	BERT	79.81	50.19	54.29	28.62	29.70	17.17	15.31	9.36	5.10	2.15	78.24	71.59	60.66	46.57
LAVT [5]	Swin-B	BERT	80.97	51.84	58.70	30.27	31.09	17.34	15.55	9.52	4.64	2.09	**78.50**	71.86	61.53	47.40
CroBIM	Swin-B	BERT	87.24	64.83	75.17	44.41	44.78	17.28	19.03	9.69	**6.26**	2.20	**78.85**	72.30	65.79	52.69
CroBIM	ConvNeXt-B	BERT	**93.04**	**75.89**	**87.70**	**61.42**	**66.13**	**34.07**	**26.91**	**12.99**	5.80	**2.75**	77.95	**72.33**	**71.93**	**59.77**
CroBIM-U	ConvNeXt-B	BERT	**94.11**	**76.68**	**88.54**	**62.54**	**68.45**	**34.81**	**27.22**	**13.50**	**6.57**	**3.26**	78.03	**73.81**	**72.55**	**60.08**

thresholds (e.g., Pr@0.7–Pr@0.9) and IoU-based scores are particularly indicative of a model's capability in accurate boundary delineation.

As shown in Table IV, CroBIM-U achieves the best performance across all precision thresholds. On the test set, it attains 76.68% (Pr@0.5) and 62.54% (Pr@0.6), surpassing previous strong competitors such as RIS-DMMI [55] (63.89%/44.30%). More importantly, CroBIM-U maintains a clear advantage under more stringent criteria that emphasize boundary quality: it reaches 34.81% (Pr@0.7), 13.50% (Pr@0.8), and 3.26% (Pr@0.9) on the test split. These gains under high overlap thresholds demonstrate that CroBIM-U produces masks with tighter alignment to fine object contours, which is crucial for this high-resolution dataset.

CroBIM-U also delivers the best IoU-based results, achieving 73.81% oIoU and 60.08% mIoU on the test set, indicating improved overall mask completeness and robustness in complex scenes. Compared with CroBIM using the same ConvNeXt-B backbone, CroBIM-U further improves the test performance from 75.89% to 76.68% on Pr@0.5 and from 61.42% to 62.54% on Pr@0.6, while boosting oIoU by +1.48%. These consistent improvements, especially under

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TABLE V

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RISBENCH DATASET. OPTIMAL AND SUBOPTIMAL PERFORMANCE IN EACH METRIC ARE MARKED BY RED AND BLUE, RESPECTIVELY

Method	Visual Encoder	Text Encoder	Pr@0.5		Pr@0.6		Pr@0.7		Pr@0.8		Pr@0.9		oIoU		mIoU	
			Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
RRN [54]	ResNet-101	LSTM	54.62	55.04	46.88	47.31	39.57	39.86	32.64	32.58	11.57	13.24	47.28	49.67	42.65	43.18
LSCM [58]	ResNet-101	LSTM	55.87	55.26	47.24	47.14	40.22	40.10	33.55	33.29	12.78	13.91	47.99	50.08	43.21	43.69
BRINet [59]	ResNet-101	LSTM	52.11	52.87	45.17	45.39	37.98	38.64	30.88	30.79	10.28	11.86	46.27	48.73	41.54	42.91
MAttNet [64]	ResNet-101	LSTM	56.77	56.83	48.51	48.02	41.53	41.75	34.33	34.18	13.84	15.26	48.66	51.24	44.28	45.71
CMPC [15]	ResNet-101	LSTM	54.89	55.17	47.77	47.84	40.38	40.28	32.89	32.87	12.63	14.55	47.59	50.24	42.83	43.82
CMPC+ [60]	ResNet-101	LSTM	57.84	58.02	49.24	49.00	42.34	42.53	35.77	35.26	14.55	17.88	50.29	53.98	45.81	46.73
LAVT [5]	Swin-B	BERT	68.27	69.40	62.71	63.66	54.46	56.10	43.13	44.95	21.61	25.21	69.39	74.15	60.45	61.93
RMSIN [4]	Swin-B	BERT	70.05	71.01	64.64	65.46	56.37	57.69	44.14	45.50	21.40	25.92	69.51	74.09	61.78	63.07
LGCE [1]	Swin-B	BERT	68.20	69.64	62.91	64.07	55.01	56.26	43.38	44.92	21.58	25.74	68.81	73.87	60.44	62.13
CrossVLT [63]	Swin-B	BERT	70.05	70.62	64.29	65.05	56.97	57.40	44.49	45.80	21.47	26.10	69.77	74.33	61.54	62.84
CARIS [56]	Swin-B	BERT	73.46	73.94	68.51	68.93	60.92	62.08	48.47	50.31	24.98	29.08	**70.55**	**75.10**	64.40	65.79
RIS-DMMI [55]	Swin-B	BERT	71.27	72.05	66.02	66.48	58.22	59.07	45.57	47.16	22.43	26.57	**70.58**	**74.82**	62.62	63.93
robust-ref-seg [23]	Swin-B	BERT	67.42	69.15	61.72	63.24	53.64	55.33	40.71	43.27	19.43	24.20	69.50	74.23	59.37	61.25
CroBIM	Swin-B	BERT	76.59	75.75	71.73	70.34	64.32	63.12	53.18	51.12	28.53	28.45	69.08	73.61	67.52	67.32
CroBIM	ConvNeXt-B	BERT	**77.41**	**77.55**	**72.62**	**72.83**	**66.74**	**66.38**	**55.92**	**55.93**	**32.17**	**34.07**	69.12	73.04	**68.70**	**69.33**
CroBIM-U	ConvNeXt-B	BERT	**77.89**	**77.73**	**72.75**	**73.40**	**67.90**	**67.05**	**57.12**	**56.85**	**33.33**	**35.52**	69.12	73.04	**69.15**	**69.62**

strict overlap thresholds, confirm the superiority of CroBIM-U in fine-grained boundary delineation on RefSegRS in Fig. 4.

3) RISBench: We conduct experiments on the RISBench dataset and report the results on both the validation and test sets in Table V. As a large-scale benchmark with **extreme scale variations** (spatial resolutions ranging from 0.1 to 30 m) [2], RISBench poses substantial challenges to referring segmentation models: targets may appear with drastically different sizes and visual patterns across observation altitudes, making robust multiscale localization and precise mask prediction particularly critical.

As shown in Table V, CroBIM-U achieves the best performance across all precision thresholds (Pr@0.5–Pr@0.9) and delivers the highest mIoU on both splits. On the test set, CroBIM-U reaches 77.73% (Pr@0.5) and 73.40% (Pr@0.6), exceeding strong BERT-based competitors such as CARIS [56] (73.94%/68.93%) and CrossVLT [63] (70.62%/65.05%). More importantly, under stricter overlap constraints that better reflect fine localization quality across varying object scales, CroBIM-U maintains clear advantages, achieving 67.05% (Pr@0.7), 56.85% (Pr@0.8), and 35.52% (Pr@0.9) on the test set. The improvements at high thresholds indicate that CroBIM-U is more capable of producing tightly aligned masks even when targets are small, thin, or visually ambiguous due to resolution changes.

In terms of overall segmentation quality, CroBIM-U obtains the highest mIoU of 69.15% (Val) and 69.62% (Test), demonstrating stronger category-balanced performance on this large-scale dataset with substantial size diversity. Compared with CroBIM using the same ConvNeXt-B backbone, CroBIM-U consistently improves high-threshold precision (e.g., +1.45% on Pr@0.9 for the test split) and further boosts mIoU (+0.45% Val/+0.29% Test), validating that resolving referential uncertainty is beneficial when targets exhibit large scale fluctuations and similar instances may coexist across different resolutions.

TABLE VI

ABLATION RESULTS ON RISBENCH (TEST SPLIT)

Setting	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
Base	77.55	66.38	34.07	**73.04**	69.33
+UGF	77.68	66.72	34.45	73.03	69.44
+UDLR	77.60	66.93	35.10	73.02	69.53
Full	**77.73**	**67.05**	**35.52**	**73.04**	**69.62**

Notably, CroBIM-U shows comparable oIoU to CroBIM on RISBench, while significantly improving mIoU and high-threshold precision. This is expected because oIoU is more influenced by large-area regions, whereas RISBench contains many small-to-medium targets whose accurate segmentation is better reflected by mIoU and stricter Pr metrics. Overall, these results confirm that CroBIM-U offers stronger robustness to scale variation and yields more precise pixel-level localization on RISBench in Fig. 5.

V. ABLATION STUDY

To assess the effectiveness of our designs within the CroBIM framework, we conduct comprehensive ablation studies on the RISBench dataset and report the results on the test split. Unless otherwise specified, all module-wise ablations share the same backbone (ConvNeXt-B with SMLFR pretraining), training schedule, and evaluation protocol and differ only in whether the proposed modules are enabled. We start from the base model (CroBIM) and progressively activate the uncertainty-driven components to form: 1) base (CroBIM); 2) +UGF; 3) +UDLR; and 4) full (CroBIM-U). The quantitative comparison is summarized in Table VI, and we further provide qualitative analysis to interpret how the learned uncertainty prior guides selective fusion and focused local correction. Besides the module-wise ablations in Table VI, we further

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TABLE VII
GENERALIZATION ANALYSIS ACROSS BACKBONES ON RISBENCH (TEST SPLIT)

Backbone	Method	Pr@0.9	mIoU
ConvNeXt-B	Base	34.07	69.33
ConvNeXt-B	Full	35.52	69.62
Swin-B	Base	33.70	68.10
Swin-B	Full	34.60	68.80

TABLE VIII
ABLATION ON THE WARM-UP RATIO FOR ENABLING $\mathcal{L}_{\text{unc}}$ (RISBENCH TEST SPLIT, FULL MODEL)

Warm-up	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
0%	75.12	63.48	31.25	70.33	66.84
5%	76.95	66.12	34.68	72.15	68.73
10% (Def)	77.73	67.05	35.52	73.04	69.62
20%	77.41	66.58	35.02	72.61	69.15

analyze the warm-up duration for enabling $\mathcal{L}_{\text{unc}}$, perform a 2-D sensitivity study over the UDLR hyperparameters (τ, t), quantify the computational overhead of each configuration, and evaluate generalization across different backbones.

A. Generalization Across Backbones

To verify that the performance gains of CroBIM-U are not tied to a specific backbone or specialized pretraining, we further evaluate our method using a Swin-Transformer backbone (Swin-B) with standard ImageNet-1K pretrained weights, which has distinct architectural inductive biases from ConvNeXt. Despite Swin-B exhibiting a lower baseline performance due to the domain gap, CroBIM-U consistently improves performance and achieves a notable +0.90% gain on the strict Pr@0.9 metric as shown in Table VII, supporting the plug-and-play, backbone-agnostic nature of our uncertainty-driven design.

B. Effectiveness of RUS

We first analyze the RUS qualitatively, since it provides the key intermediate representation—the pixel-wise uncertainty map U—that subsequently guides both UGF and UDLR. As described in Section III-C, U is learned with an online error-aligned supervision signal, encouraging it to respond to locations that are prone to referential confusion or local segmentation errors.

Since the online error-aligned supervision can be noisy in the early training stage, we enable $\mathcal{L}_{\text{unc}}$ only after a short warm-up. Table VIII varies the warm-up ratio on RISBench (test split) using the Full model and shows that a 10% warm-up provides the best overall performance.

Although RUS predicts uncertainty on stride-32 tokens (for efficiency and a large receptive field), the dominant uncertainty in referring remote-sensing segmentation is often *semantic ambiguity* (e.g., multiple visually similar candidates under the same expression) rather than purely high-frequency edge noise. Therefore, a coarse uncertainty prior can reliably indicate risky regions for activating UGF/UDLR, while fine-grained boundary recovery is handled by UDLR using higher-resolution decoder features.

As illustrated in Fig. 6, each row presents a representative case with the referring expression, original image, GT, prediction, the predicted uncertainty map, and a zoomed-in detail view. The three rows correspond to different sources of uncertainty in remote-sensing referring segmentation.

1) *Row 1 Dense Scenes With Similar Instances:* The uncertainty map shows strong responses around multiple visually similar candidates, reflecting increased referential ambiguity when targets are surrounded by distractors.
2) *Row 2 Small-Scale Targets:* The uncertainty becomes higher near the referred small object (e.g., a car) due to limited visual evidence and weak saliency, making localization and segmentation more fragile.
3) *Row 3 Boundary- and Structure-Sensitive Regions:* For elongated targets with thin boundaries (e.g., the harbor), uncertainty concentrates along edges and local transition areas, where slight geometric deviations can cause boundary shifts, breakage, or adhesion.

These observations indicate that U captures a spatial risk prior aligned with different failure modes, rather than a generic saliency cue, and thus provides reliable guidance for UGF and UDLR.

C. Effectiveness of UGF

We then evaluate the contribution of UGF, which uses the uncertainty map to modulate the strength of language injection. UGF is designed to address the dual effect of language cues in remote-sensing scenes: while textual constraints are essential for disambiguation among visually similar instances, uniformly injecting language information can introduce unnecessary perturbations in already confident regions and may even cause referring drift. By gating fusion strength with U, UGF strengthens cross-modal constraints in high-uncertainty regions and suppresses ineffective perturbations in low-uncertainty regions.

The ablation results in Table VI show that enabling RUS+UGF consistently improves segmentation performance over the base model, with the most noticeable gains appearing in Pr@0.5 (and typically Pr@0.7). This trend aligns with the intended role of UGF: it primarily improves instance-level grounding reliability by reducing confusion among candidate instances under complex descriptions, thereby increasing the probability that the predicted mask overlaps with the correct target at moderate thresholds. Meanwhile, the improvement on Pr@0.9 is relatively smaller, suggesting that UGF mainly affects correct localization and global alignment rather than directly refining fine boundaries.

D. Effectiveness of UDLR

Finally, we assess UDLR, which performs lightweight residual refinement guided by an uncertainty-derived soft region

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TABLE IX
2-D SENSITIVITY ANALYSIS OF UDLR HYPERPARAMETERS (τ, t) (MIOU ON RISBENCH TEST SPLIT, FULL MODEL)

$\tau \backslash t$	0.08	0.10 (Def)	0.15
0.40	69.05	69.33	69.48
0.50	69.54	69.62	69.57
0.60	69.22	69.49	69.36

TABLE X
COMPUTATIONAL OVERHEAD ANALYSIS (INPUT 512 × 512, FP32, MEASURED ON A800)

Configuration	Params	Δ	GFLOPs	Δ	Latency	Δ
Base	358.7M	0.00%	145.3	0.00%	472ms	0.00%
Full	359.5M	+0.22%	158.6	+9.19%	531ms	+12.52%
RUS+UGF	359.0M	+0.09%	147.7	+1.67%	487ms	+3.13%
RUS+UDLR	359.1M	+0.12%	156.2	+7.52%	505ms	+6.92%

mask. Remote-sensing targets frequently contain thin, fragmented, or low-contrast structures, and the dominant errors tend to cluster around boundaries and fine details, manifesting as merging/bleeding, discontinuities, and boundary shifts [1]. UDLR explicitly addresses these issues by focusing corrections on high-risk locations indicated by U and avoiding unnecessary updates in confident regions.

As reported in Table VI, enabling RUS+UDLR yields consistent improvements over the base model, with larger gains on Pr@0.9 and mIoU. This behavior matches the design objective of UDLR: high-threshold precision is especially sensitive to boundary alignment and shape fidelity, and mIoU better reflects quality improvements on diverse targets (including smaller ones) that are common in remote-sensing imagery. In contrast, the change in oIoU is typically marginal, which is expected because oIoU is more influenced by large-area regions, whereas UDLR focuses on refinement on localized boundary and thin-structure areas.

We conduct a 2-D sensitivity study over the UDLR hyperparameters (τ, t) on RISBench (test split) using the Full model. As shown in Table IX, the performance varies only slightly within a reasonable range, indicating that UDLR is not brittle to specific tuning.

E. Complementarity of UGF and UDLR

We further combine UGF and UDLR to form the full risk-guided fusion–local correction pipeline. The Full model achieves the best overall performance, demonstrating that UGF and UDLR are complementary: UGF improves cross-modal disambiguation and stabilizes global grounding, while UDLR enhances geometric fidelity by correcting boundary and thin-structure errors. Together, they validate the effectiveness of using an error-aligned uncertainty before guiding both selective fusion and focused local refinement in remote-sensing referring segmentation.

F. Computation Overhead

We further quantify the computational overhead of each configuration to support the "lightweight" claim. Table X reports Params, GFLOPs, and latency (input 512 × 512, FP32, measured on A800). Overall, the Full model introduces only a marginal parameter increase and moderate latency overhead, while RUS+UGF remains highly efficient.

VI. CONCLUSION

In this work, we proposed an uncertainty-driven framework for remote-sensing referring segmentation, motivated by the spatially nonuniform reliability of cross-modal grounding and pixel-wise prediction in complex remote-sensing scenes. We introduced a pixel-wise cross-modal referential uncertainty map as a unified spatial prior and developed an online error-aligned risk learning strategy to make the predicted uncertainty interpretable and consistent with actual model errors. Building on this prior, we designed three plug-and-play modules—RUS, UGF, and UDLR—to enable uncertainty-guided language modulation and focused local correction, strengthening disambiguation in high-risk regions while preserving stable predictions in confident areas.

Extensive experiments on three remote-sensing benchmarks demonstrate that our method consistently improves segmentation accuracy and boundary/detail quality, achieving state-of-the-art performance without modifying the backbone architecture. In future work, we will explore more fine-grained uncertainty modeling and extend the proposed risk-aware paradigm to broader vision–language geospatial understanding tasks.

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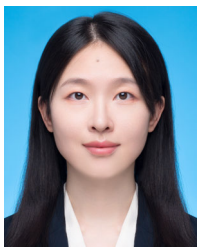
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